API TITLE*1995 INT ** ■ 0732290 0550476 52T ■

1994 and 1995 Interpretations—Technical Publications

Answers to Technical Inquiries

FEBRUARY 1996



American Petroleum Institute

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API TITLE*1995 INT ** 0732290 0550477 466

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Answers to Technical Inquiries

Manufacturing, Distribution and Marketing Department

FEBRUARY 1996

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FOREWORD

This publication is a compilation of interpretations issued by the Industry Operattions Group of the American Petroleum Institute. API has issued these interpretations in response to questions concerning the meaning of its standards and in consideration of suggestions for revisions to its standards.

API welcomes questions, suggestions, and comments concerning its standards. However, API can only respond to questions concerning application of its standards that are stated clearly and that strictly pertain to the meaning of one of its standards. API staff will make interpretations of standards in writing after consultation, if necessary, with the appropriate committee officers and the committee membership.

Suggestions for revision to an API standard should include supporting technical information. Inquiries that do not provide the information needed for complete understanding will be returned.

As a matter of policy, API does not approve, certify, rate, or endorse any item, contribution, proprietary device, or activity. Moreover, API does not act as a consultant on specific engineering problems or on the general understanding or application of its standards. Therefore, inquiries requesting such considerations will be returned.

API does not normally respond to questions seeking the rationale for requirements in its standards. These requirements are based upon consideration of technical data and the judgment and skill of experienced engineering and technical personnel, representing both users and manufacturers, who serve on API standards committees.

API committees meet regularly to consider written requests for interpretations and revisions of standards, as well as to develop new standards as dictated by technological development. All technical meetings during which requirements of API standards are considered are open to the public. Notices of these meetings are published in *Standards Action*, a publication of the American National Standards Institute. [1]A report of standardization activities is also available from individual API departments.

API publications may be used by anyone desiring to do so. Every effort has been made by the Institute to assure the accuracy and reliability of the data contained in them; however, the Institute makes no representation, warranty, or guarantee in connection with this publication and hereby expressly disclaims any liability or responsibility for loss or damage resulting from its use or for the violation of any federal, state, or municipal regulation with which this publication may conflict.

[1]American National Standards Institute, 1430 Broadway, New York, New York 10018.

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HOW TO PREPARE A TECHNICAL INQUIRY

Inquiries sent to the American Petroleum Institute should either request an interpretation of an API standard or ask that a revision to an API standard be considered on the basis of new data or technology. The following guidelines are recommended for submitting inquiries:

1. Format: Include your name and mailing address on all correspondence. Your letter should be typewritten (although legible handwritten letters will be considered), and mailed to:

American Petroleum Institute
1220 L Street, Northwest
Washington, D.C. 20005

2. Scope: Limit your inquiry to a single subject or closely related subjects. An inquiry letter concerning unrelated subjects will be returned.
3. Background: State the purpose of your inquiry and your understanding of the standard's requirements. Provide all information needed for a complete understanding of your inquiry. Refer to the applicable edition, revision, paragraphs, figures, and tables. Include sketches, if needed.
4. Inquiry: State your inquiry in a condensed, precise question format, omitting any superfluous background information. The question itself should be technically and editorially correct. Try to compose your question in such a way that "yes" or "no," with provisos if needed, would be a suitable reply. For example, you could ask whether or not your understanding of a standard's requirements is correct.
5. Suggestion for a Revision: When proposing a revision to an API standard, you should include supportive technical reasons and data, along with proposed new wording. Comments made concerning the technical basis for a requirement of an API standard should include relevant notes and suggestions. These suggestions and comments will be forwarded to the appropriate API subcommittee for the next review of that standard.

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API TITLE*1995 INT ** 0732290 0550482 823

Issue Date: July 6, 1994

Publication: API Specification 5B, *Specification for Threading, Gaging and Thread Inspection of Casing, Tubing, and Line Pipe Threads*

Edition/Date: Thirteenth Edition, May 31, 1988

Question: Is the taper measurement interval 1 inch for 2⅜ EUE 8 round tubing in API Specification 5B?

Reply: I agree that the measurement interval is 1 inch. However, the taper measurement shall include all threads between the first and last perfect thread.

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API TITLE*1995 INT ** 0732290 0550483 76T

Issue Date: May 25, 1995

Publication: API Specification 5B, *Specification for Threading, Gaging and Thread Inspection of Casing, Tubing, and Line Pipe Threads*

Edition/Date: Thirteenth Edition, May 31, 1988

Question 1: Are the long thread external-upset tubing tolerances in Table 2.10 of API Specification 5B required for API Specification 15LR line pipe with API Specification 5B, Table 2.6a long threads?

Reply 1: No. Long thread external-upset tubing thread tolerances are not required for line pipe threads.

Question 2: Will pipe made to API Specification 5B line pipe thread tolerances work in an application where it is interchanged or made-up with pipe threaded to tighter tubing tolerances?

Reply 2: Applications are not covered by the standards; however, I am unaware of any application problems relating to pipe of a given thread size with a smaller tubing thread tolerance, which is made-up to pipe of the same thread size with the larger line pipe thread tolerance. I understand that it is common field practice to connect API fiberglass tubing threads to API steel line pipe threads of a given size.

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API TITLE*1995 INT ** 0732290 0550484 6T6

Issue Date: May 25, 1994

Publication: API Specification 5CT, *Specification for Casing and Tubing*

Edition/Date: Fourth Edition, November 1, 1992

Question 1: When a processor applies an "in-process" serial number, die stamped on the pipe end (face), which will be removed during subsequent threading, and then includes this stamped number into the stencil at a later time, is this in compliance with the requirements of Paragraph D.4.b.4 in Appendix D of the Fourth Edition of API Specification 5CT?

Reply 1: Yes, provided the processor's documented procedures can demonstrate that traceability is maintained.

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API TITLE*1995 INT ** ■ 0732290 0550485 532 ■

Issue Date: March 20, 1995

Publication: API Specification 5CT, *Specification for Casing and Tubing*

Edition/Date: Fourth Edition, April 1, 1995

Question 1: Do wall thickness measurements using radiation thickness gaging equipment that calculate an average wall thickness satisfy the specification requirements?

Reply 1: Paragraph 9.7.4 of the Fifth Edition of API Specification 5CT states "All seamless casing, tubing, liners, and pup joints requiring electromagnetic or ultrasonic inspection as specified in Table 55 shall have the wall thickness verified in a helical or longitudinal path over the length of the pipe, excluding end areas..."

I interpret these words to mean that this inspection system is acceptable provided that the manufacturer demonstrates the capability of the inspection system to verify that the wall thickness requirement of Paragraph 7.3 has been met in a path which will take into account the eccentricity of the seamless process over the full length, except specified end areas.

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API TITLE*1995 INT ** 0732290 0550486 479

Issue Date: May 13, 1994

Publication: API Specification 5L, *Specification for Line Pipe*

Edition/Date: Fortieth Edition, November 1, 1992

Question: Is a manufacturer in violation of the API Specification 5L Paragraph 9.14 requirement: "The equipment ... shall be capable of inspecting 1/16-inch on either side of the weld line for the entire wall thickness" if the manufacturer's ultrasonic equipment is designed and calibrated to alarm on weld line defects only and does not expect that equipment to alarm on defects that could be found by visual inspection?

Reply: Yes. It is not apparent that this practice is in compliance with Paragraph 9.14 of API Specification 5L "unless it can be demonstrated by the manufacturer that the imperfection does not exceed the provisions of Paragraph 10.3" per API Specification 5L Paragraph 9.16. (As you know, Paragraph 10.3 includes defect criteria.)

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API TITLE*1995 INT ** 0732290 0550487 305

Issue Date: September 13, 1994

Publication: API Specification 5L, *Specification for Line Pipe*

Edition/Date: Fortieth Edition, November 1, 1992

Question: Is seam welding a special process for API Specification 5L double submerged-arc welded (DSAW) pipe?

Reply: Yes. The API Specification Q1 requirement for qualification of special processes such as welding results from the fact that all material properties of the weld and heat-affected zone cannot be fully verified by subsequent inspection and testing.

Although this issue is under review by API standards committees, welding is currently a mandatory special process for all API Specification 5L products as shown in Paragraph 1.6 of the fortieth edition.

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API TITLE*1995 INT ** 0732290 0550488 241

Issue Date: March 21, 1995

Publication: API Specification 5L, *Specification for Line Pipe*

Edition/Date: Fortieth Edition, November 1, 1992

Question: Is line pipe with continuous roll marks of up to 8 percent of the specified wall thickness in compliance with the workmanship requirements of Paragraph 10.3 in the fortieth edition of API Specification 5L?

Reply: Since the depth of the roll marks does not violate API Specification 5L defect requirements, acceptance of the roll marks under the workmanship provisions in Paragraph 10.3 is subject to agreement between the purchaser and manufacturer.

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API TITLE*1995 INT ** 0732290 0550489 188

Issue Date: June 13, 1995

Publication: API Specification 5L, *Specification for Line Pipe*

Edition/Date: Forty-First Edition, April 1, 1995

Question: Is the use of a gage fabricated to measure the 1/16-inch (1.59 mm) linear gap acceptable when checking the squareness of API Specification 5L pipe ends inspected at the mill by an angular conversion method?

Reply: Yes. The linear tolerance may be converted to an angular measurement; however, since the angular tolerance varies with the pipe diameter, the 1/16-inch maximum linear tolerance is the final acceptable criteria.

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API TITLE*1995 INT ** ■ 0732290 0550490 9TT ■

Issue Date: August 23, 1994

Publication: API Recommended Practice 6AR, *Recommended Practice for Repair and Remanufacture of Wellhead and Christmas Tree Equipment*

Edition/Date: First Edition, April 15, 1994

Question 1: Paragraphs 501.2f(2) and 601.2f(2) state that the hydrostatic body test pressure (reassembled equipment) should be determined by the rated working pressure of the equipment as defined in API Specification 6A. API Specification 6A defines rated working pressure as follows: The maximum internal pressure equipment is designed to contain and/or control. Working pressure is not to be confused with test pressure. Please clarify the referenced paragraphs.

Reply 1: The hydrostatic test pressures are determined by the rated working pressure. The hydrostatic body test for assembled equipment is to be performed at the pressures listed in Table 605.10 of API Specification 6A.

Question 2: Since Paragraphs 501 and 601 do not recommend hydrostatic testing of individual components after disassembly, should one assume that hydrostatic testing need only be performed on the completed reassembly — i.e., christmas trees and frac heads unless otherwise requested by the user?

Reply 2: Paragraph 501.2(f) and 601.2(f) are titled "Equipment Testing." API Specification 6A defines equipment as "any single completed unit that can be used for its intended purpose without further processing on assembly (e.g., a valve, choke, cross, tee, spool, etc.)." Thus, the testing provisions of Paragraph 501.2(f) and 601.2(f) apply to individual pieces of equipment.

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API TITLE*1995 INT ** [redacted] 0732290 0550491 836 [redacted]

Issue Date: August 9, 1995

Publication: API Specification 7, *Specification for Rotary Drill Stem Elements*

Edition/Date: Thirty-Eighth Edition, April 1, 1994

Question 1: Is the radius undercut of the stress-relief groove on the pin shoulder of the referenced* drawing in compliance with API Specification 7?

Reply 1: Yes. API Specification 7 does not require that the make up shoulder be tangent to the ¼-inch radius at the base of the pin in Figure 6.2.

Question 2: If this undercut is made, would it then become a non-API connection?

Reply 2: No.

Question 3: If this undercut is not made, would a step on the shoulder resulting from refacing after use be a stress riser?

Reply 3: This issue is outside the scope of API specifications and recommended practices.

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API TITLE*1995 INT ** ■ 0732290 0550492 772 ■

Issue Date: December 1, 1995

Publication: API Recommended Practice 7G, *Recommended Practice for Drill Stem Design and Operating Limits*

Edition/Date: Fifteenth Edition, January 1, 1995

Question: In the slip area of used drill pipe tubes, can cuts and gouges which have a depth greater than 10 percent of the average adjacent wall be ground out if the remaining wall is not reduced below 80 percent of the specified wall thickness and still meet premium class requirements? (Reference Table 19, API Recommended Practice 7G, Fifteenth Edition.)

Reply: Yes, provided the cut or gauge is completely removed.

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API TITLE*1995 INT ** ■ 0732290 0550493 609 ■

Issue Date: May 27, 1994

Publication: API Specification 13A, *Specification for Drilling Fluid Materials*

Edition/Date: Fifteenth Edition, May 1, 1993

Question: Is a Hamilton Beach Model 941 three Spindle Mixer with factory rpm setting of 13,000, 16,000, and 18,000 reduced by means of a voltage reducer to 11,500 rpm acceptable for Q1? My concern is the variation of under load speeds of this Hamilton Beach versus Multimixer Model 9B. I would appreciate an interpretation of this.

Reply: API Specification 13A, Fifteenth Edition dated May 1, 1993, does not preclude you from using a Hamilton Beach Model 941 mixer; the specification cites the Multimixer Model 9B only as an example. In drafting the specification, the Hamilton Beach mixer was considered; however, due to it variable speed under load, this type of mixer was not specifically referenced in the document. The specification does not address mixer selection but does specify the requirements that must be met for mixer calibration. Regarding mixer selection, API can only suggest that you use good business and engineering discretion in selecting your laboratory equipment.

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API TITLE*1995 INT ** 0732290 0550494 545

Issue Date: September 20, 1995

Publication: API Specification 14D, *Specification for Wellhead Surface Safety Valves and Underwater Safety Valves for Offshore Service*

Edition/Date: Ninth Edition, June 1, 1994

Question: As per the API Specification 14D, for Class 1 service, the valve is to be tested using water as the test medium. For verification testing, a valve 2¼-inch size is to be leak-tested with water at 5000 psi pressure and with Nitrogen at 2000 psi. Then water is to be circulated through the valve at a flow rate of 85 GPM for 50 hours. Subsequently, the valve is to be leak-tested again with water at 5000 psi and Nitrogen at 2000 psi. Then the valve is to be tested for cyclic testing with the valve open and close 500 times while circulating water at 85 GPM and on closure cycle, the valve maintains a 400 psi pressure differential. Then the valve is to be leak-tested in the water at 5000 psi and nitrogen at 2000 psi.

If all these tests are to be carried out in one setting, the whole circuit has to be designed for 5000 psi. Instead, we propose to carry out the leak tests separately using a hydrostatic pump of 5000 psi rating and then flow test and cyclic testing to be carried out in a circuit with a pump of closed head pressure 400 psi and nominal flow rate 85 GPM. With this the valve needs to be dismantled for carrying out leak tests after the flowing test and again the valve needs to be mounted for carrying out cyclic testing.

We would like to know whether the above will meet the requirements of API Specification 14D.

Reply: API Specification 14D states that a valve must be tested in accordance with Sections 905, 906 and 907 to be verified as a Class 1 valve. This means that in order to meet the requirements of a Class 1, API 14D valve, one test valve must be tested as specified in these sections, and pass the tests, without being modified, altered or repaired during the tests.

API is not in the business of specifying the exact methodology or apparatus used to perform this testing. The manufacturer is free to use whatever piping and testing equipment he chooses so long as the test value is tested according to the test conditions and procedures specified in API Specification 14D for Class 1 service.

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API TITLE*1995 INT ** ■ 0732290 0550495 481 ■

Issue Date: September 26, 1994

Publication: API Specification 15LR, *Specification for Low Pressure Fiberglass Line Pipe*

Edition/Date: Sixth Edition, September 1, 1990

Question: Does API Specification 15LR require fiberglass pipe manufacturers to fabricate their product under cyclic pressure conditions or does API Specification 15LR require fabrication under static pressure conditions?

Reply: According to API Specification 15LR a fiberglass pipe manufacturer may fabricate product based on cyclic pressure conditions or static pressure conditions. One should refer to Section 1, Scope, Paragraph 1.1 of that document, that addresses "static pressure conditions higher pressure ratings may be used under Paragraph 4.5(b)."

Paragraph 4.5(b) of API Specification 15LR provides a choice to the manufacturer by utilization of one of two formulas to determine the static pressure rating, whereas the static pressure rating may be calculated at 1.5 times the cyclic pressure rating.

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API TITLE*1995 INT ** 0732290 0550496 318

Issue Date: December 28, 1994

Publication: API Specification 15LR, *Specification for Low Pressure Fiberglass Line Pipe*

Edition/Date: Sixth Edition, September 1, 1990

Question: Does API Specification 15LR allow a static pressure rating up to 1500 psi?

Reply: Yes. Paragraph 1.1 notes: "For static pressure conditions higher pressure ratings may be used [refer to Paragraph 4.5(b)]."

Paragraph 4.5(b) allows a static pressure rating of 1.5 times the cyclic pressure rating, which has a maximum of 1000 psi.

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API TITLE*1995 INT ** 0732290 0550497 254

Issue Date: May 25, 1995

Publication: API Specification 15LR, *Specification for Low Pressure Fiberglass Line Pipe*

Edition/Date: Sixth Edition, September 1, 1990

Question: Are the long thread external-upset tubing tolerances in Table 2.10 of API Specification 5B required for API Specification 15LR line pipe with API Specification 5B, Table 2.6a long threads?

Reply: No. Long thread external-upset tubing thread tolerances are not required for line pipe threads.

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API TITLE*1995 INT ** 0732290 0550498 190

Issue Date: June 1, 1995

Publication: API Standard 510, *Pressure Vessel Inspection Code: Maintenance Inspection, Rating, Repair, and Alteration*

Edition/Date: Seventh Edition, March 1992

Question 1: To what extent should calculations be performed to show an acceptable increase in MAWP? In other words, is a full design verification calculation required, or just head and shell membrane stress calculations?

Reply 1: Engineering evaluation and calculations should be performed to the extent required to assure safety under the intended operating conditions.

Question 2: It appears that 5.3 (e) is subjective, and depends upon the requirements of individual API authorized inspectors. Is there an itemized checklist or written acceptance criteria used by inspectors?

Reply 2: There is no standard itemized checklist used by API Standard 510 certified inspectors. The inspector has the responsibility of assuring that all local, state, federal, and company requirements are satisfied to the best of his ability. The criteria may vary based on factors such as the intended service of the equipment in question and jurisdictional and company requirements.

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API TITLE*1995 INT ** ■ 0732290 0550499 027 ■

Issue Date: November 22, 1995

Publication: API Standard 510, *Pressure Vessel Inspection Code: Maintenance Inspection, Rating, Repair, and Alteration*

Edition/Date: Seventh Edition, March 1992

Question 1: Are API Standard 510 repairs, alterations, or rerates recognized in federal jurisdictions?

Reply 1: The API Subcommittee on Inspection, Task Group on Inspection Codes does not determine or write standards for acceptance of repairs, alterations, or rerates for federal jurisdictions. Your question may be better directed to federal personnel with that responsibility. No restrictions exist in API Standard 510 that prevent it from being used in federal jurisdictions. The forward of API Standard 510 clearly states: "API publications may be used by anyone wishing to do so."

Question 2: Is it required to send documentation of an API Standard 510 repair to the governing jurisdiction?

Reply 2: API Standard 510 does not require that documentation of repairs be sent to the "governing jurisdiction." However, jurisdictional authorities may require such documentation as a supplemental requirement in addition to the requirements of API Standard 510. The quality assurance inspection manual described in Paragraph 2.3(g) of API Standard 510, Supplement 1, requires: "Ensuring all jurisdictional requirements for pressure vessel inspection, repairs, alterations, and rerating are continuously met."

Question 3: Who's responsibility is it to send said documentation to the jurisdiction?

Reply 3: The method of transmitting the documentation to the jurisdiction, and the personnel responsible for transmitting the documentation to the jurisdiction is a matter to be agreed upon between the Owner/User organization and the jurisdiction. Requirements should be described in the quality assurance inspection manual referenced in Paragraph 2.3 of API Standard 510, Supplement 1.

Question 4: Assume an API Standard 510 repair is made to an ASME code item and all is in order. The repair organization informs an API authorized pressure vessel inspector. The repair is accepted, work is completed and the documentation is signed with the required signatures; however, the documentation is not sent to the governing jurisdiction. Is this an accepted API Standard 510 repair to the said standard?

Reply 4: Yes.

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API TITLE*1995 INT ** 0732290 0550500 679

Question 5: Is API documentation audited? By whom?

Reply 5: Yes. API Standard 510 Paragraph 2.3(e) in Supplement 1, states that the quality assurance inspection manual shall include: "Internal auditing for compliance with the quality assurance inspection manual."

Question 6: Is the National Board Commissioned Inspector with an "A" endorsement certified to perform the duties of an API Standard 510 authorized pressure vessel inspector?

Reply 6: No, however, inspectors currently qualified by written examination by the National Board of Boiler and Pressure Vessel Inspectors are eligible for API Standard 510 certification upon submittal of the required documentation.

Question 7: Can API Standard 510 inspections be performed without the control of an Owner/User inspection agency?

Reply 7: No.

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API TITLE*1995 INT ** 0732290 0550501 505

Issue Date: March 7, 1995

Publication: API Standard 510, *Pressure Vessel Inspection Code: Maintenance Inspection, Rating, Repair, and Alteration*

Edition/Date: Seventh Edition, March 1992

Question 1: Should API Standard 510 be used as a governing document for maintenance of pressure relief devices?

Reply 1: Yes, for pressure relief devices on pressure vessels covered by API Standard 510 Code.

Question 2: Is there a maintenance certification for API Standard 510 or just inspection only?

Reply 2: There is no API certification process for pressure relief device maintenance.

Question 3: Are personnel who perform maintenance on pressure relief devices required to be certified to API Standard 510?

Reply 3: No, unless they are also performing inspections of pressure vessels in accordance with API Standard 510.

Question 4: The shops which repair and test the pressure relief valves for the Mt. Vernon site have a "VR" stamp from the NBIC which certifies them to work on the valves. Wouldn't you agree that this should meet the requirements laid out in API Standard 510, Section 4.5? If so, they shouldn't have to be certified to API Standard 510 also, should they?

Reply 4: API does not approve or certify pressure relief repair facilities. Section 4.5 specifies general criteria for these facilities.

Question 5: If a pressure relief device is removed from a vessel by the owner/operator and then shipped to an external shop for repair, the personnel performing the removal by definition of API Standard 510, are members of a repair organization and need the same quality control and training as the external repair shop. True or False?

Reply 5: API Standard 510 only addresses qualifications of personnel who are inspecting, servicing and setting pressure relief devices. The document does not specify requirements for personnel who remove this equipment.

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API TITLE*1995 INT ** 0732290 0550502 441

Issue Date: November 22, 1995

Publication: API Standard 510, *Pressure Vessel Inspection Code: Maintenance Inspection, Rating, Repair, and Alteration*

Edition/Date: Seventh Edition, March 1992

Question: If the Owner/User inspectors are certified to Appendix B of API Standard 510, does it mean that the independent individual or contractor require the same credentials to perform visual inspections and report findings of discontinuities?

Reply: Yes, if the Owner/User program requires that level of certification.

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API TITLE*1995 INT ** 0732290 0550503 388

Issue Date: November 28, 1995

Publication: API Standard 510, *Pressure Vessel Inspection Code: Maintenance Inspection, Rating, Repair, and Alteration*

Edition/Date: Seventh Edition, March 1992

Question: Please explain in determination of allowable working pressure, when calculating actual thickness, what does it mean to use twice the estimated corrosion loss before the date of the next inspection?

Reply: The procedure in 3.4 for determining maximum allowable working pressure (MAWP) requires subtracting twice the estimated corrosion loss before the next inspection from the thickness used in the MAWP calculation to preserve the corrosion half-life basis for setting the inspection interval. Paragraph 4.3 requires that the next inspection be made before more than half the remaining corrosion life is expended, to provide a safety margin for unexpected variation in corrosion rate.

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API TITLE*1995 INT ** ■ 0732290 0550504 214 ■

Issue Date: April 1994

Publication: API Recommended Practice 520, *Sizing, Selection, and Installation of Pressure-Relieving Devices in Refineries,* Part I, "Sizing and Selection"

Edition/Date: Sixth Edition, March 1993

Question: Regarding your question on sphere fire heat input.

Reply: Before commenting on your inquiry, it may be worth noting that the methods for determining heat input for vessels and tanks exposed to external fires are currently under review. API recognizes that different methods are described in API Recommended Practices 520 and 521 and API Standard 2000. There is an effort underway to make the technical basis for these two standards consistent however, to our knowledge, both methods have provided adequate protection from overpressure due to external fires.

Your inquiry indicated that the subject vessels are C3 and C4 storage spheres. Assuming that the design pressure is above 15 psig (i.e., non-refrigerated, based on the vapor pressure of C3's and C4's), they would fall outside the scope of API Standard 2000, which is limited to 15 psig. The appropriate method of calculating the venting requirements can be found in API Recommended Practice 520.

Another part of your inquiry state that the heat input calculated using equation 2 ($Q=34{,}500FA^{0.82}$) is twice that of equation 1 ($Q=21{,}000FA^{0.82}$). The ratio of these two equations, using the same wetted area and F factor, should be 34,500/21,000 = 1.64.

Your inquiry also referred to a 0.5 factor for good drainage, found in API Standard 2000, Paragraph 1.3.2.7. This factor, which appeared in the third edition of Standard 2000, has been further defined as "impoundment away from tank" in the current fourth edition. It now appears in Table 4.

As always, local laws and regulations should be reviewed to determine if they specify a method for estimating heat input and vapor for the fire case.

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API TITLE*1995 INT ** 0732290 0550505 150

Issue Date: April 15, 1994

Publication: API Recommended Practice 520 *Sizing, Selection, and Installation of Pressure-Relieving Devices in Refineries,* Part I, *"Sizing and Selection"*

Edition/Date: Sixth Edition, March 1993

Question: Please clarify questions regarding a typographical error and a technical inconsistency in Footnote B of Table D-3 of Recommended Practice 520, Part I (Fifth Edition).

Reply: The error you noted was with the units of thermal conductivity. You correctly noted that these units should be "Btu-in/Hr-Ft^2-°F" and the correction will be made in the next edition.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1995 INT ** ■ 0732290 0550506 097 ■

Issue Date: April 15, 1994

Publication: API Recommended Practice 520, *Sizing, Selection, and Installation of Pressure-Relieving Devices in Refineries,* Part I, "Sizing and Selection"

Edition/Date: Sixth Edition, March 1993

Question: Question regarding errors in the equations for the outlet Mach number (Recommended Practice 520, Part I) and critical pressure in Recommended Practice 521.

Reply: Your observations are correct and these equations will be corrected in the Fourth Edition of Recommended Practice 521, which will be issued during 1994.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1995 INT ** 0732290 0550507 T23

Issue Date: November 11, 1994

Publication: API Recommended Practice 520, *Sizing, Selection, and Installation of Pressure-Relieving Devices in Refineries,* Part I, "Sizing and Selection"

Edition/Date: Sixth Edition, March 1993

Question: Please clarify the flow rate to be used for calculating inlet pressure losses.

Reply: Appendix M of ASME Code, Section VIII, states in Paragraph M-7, "The inlet pressure losses will be based on the valve nameplate capacity corrected for the characteristics of the flowing fluid".

There is justification for this approach to inlet line sizing. The valve will not be called upon to relieve more than the nameplate capacity if it has been properly sized. It will only relieve whatever the relief load is for the system.

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API TITLE*1995 INT ** 0732290 0550508 96T

Issue Date: November 20, 1995

Publication: API Recommended Practice 520, *Sizing, Selection, and Installation of Pressure-Relieving Devices in Refineries,* Part II, "Installation"

Edition/Date: Fourth Edition, December 1994

Question 1: Is the 3 percent pressure loss for the inlet piping between the protected equipment and the inlet flange an approximate guideline or is it a maximum value that should never be exceeded?

Reply 1: The statements regarding the 3 percent pressure loss in inlet piping are guidelines.

Question 2: If the standard of 3 percent pressure loss is just a guideline, what would be the maximum allowable pressure loss in terms of percentage of the set pressure of the valve?

Reply 2: If a user would like to exceed the 3 percent guideline, an engineering analysis should be performed and the valve vendor should be consulted. The pressure drop allowable would depend on the outcome of the engineering analysis.

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API TITLE*1995 INT ** ■ 0732290 0550509 8T6 ■

Issue Date: April 15, 1994

Publication: API Recommended Practice 521, *Guide for Pressure-Relieving and Depressuring Systems*

Edition/Date: Third Edition, November 1990

Question: Please clarify 3.18.2 of API Recommended Practice 521 (Third Edition – 1990). The wording is confusing.

Reply: API agrees the wording in this section could be confusing. Therefore, in the fourth edition revised wording is being drafted. The proposed revised wording is as follows:

"Complete tube rupture, where a large quantity of high pressure fluid will flow to the lower pressure exchanger side is a remote contingency. Minor leakage can seldom over pressure an exchanger during operation. Since standard hydrostatic test pressure is 150 percent of the equipment design pressure, equipment failure (i.e., loss of containment of the low pressure side to the atmosphere) is unlikely where the low pressure side (including upstream and downstream systems) is designed for at least two-thirds of the design pressure of the high pressure side. Where actual test pressure of the low pressure side is less than 150 percent of the design pressure, this lower pressure should be used to determine whether pressure relief protection is needed. The use of operating pressure instead of design pressure where there is a sub-standard difference in the design pressure and the operating pressure of the high pressure side should be considered on a case-by-case basis. Pressure relief protection for tube rupture is not required where the low pressure exchanger side is designed at or above this two-thirds criteria, except as described below.

Exchanger with wide differences in the vapor pressure and temperature between the two streams should be evaluated for pressure relief protection because vapor generation and an associated pressure increase may result when the two fluids are mixed."

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API TITLE*1995 INT ** 0732290 0550510 518

Issue Date: April 15, 1994

Publication: API Recommended Practice 521, *Guide for Pressure-Relieving and Depressuring Systems*

Edition/Date: Third Edition, November 1990

Question: Please clarify the wording that defines the flow through split in API Recommended Practice 521 (Third Edition – 1990).

Reply: The language that will appear in the fourth edition should satisfy your concerns. The proposed wording is as follows:

"....For the purpose of determining the required relieving flow rate, the following basis should be used:

1. The tube failure is a sharp break in one tube.

2. The tube failure is assumed to occur at the tube to tube sheet interface.

3. The high pressure fluid is assumed to flow both through the opening in the tube sheet and through the tube at the tube sheet interface.

4. A simplifying assumption of two orifices may be used, since this will produce a larger relief flow rate than the above approach of a long open tube and open tube sheet."

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API TITLE*1995 INT ** ■ 0732290 0550511 454 ■

Issue Date: March 17, 1995

Publication: API Recommended Practice 530, *Calculation of Heater Tube Thickness in Petroleum Refineries*

Edition/Date: Third Edition, September 1988

Question: Have there been any changes in the allowable stress rupture values for A 335 P 22 in the editions of API Recommended Practice 530 after the 1972 edition?

Reply: This reply has two parts. The first deals with the question asked and the second deals with the application of the API Recommended Practice 530 design procedure.

First, there have been three editions of API Recommended Practice 530: first edition-1958; second edition-1978 and; third edition-1988 (current). The Rupture Allowable Stress curves for P 22 in the first edition were based on ASTM-ASME Special Technical Publication No. 151 (1953). The Rupture Allowable Stress curves for P 22 in the second and third editions of API Recommended Practice 530 were based on ASTM Data Series Publication 6S2 (1971). Because the data sources were different, there are differences between the Rupture Allowable Stress curves in the first and second editions. The curves in the second and third editions are the same, however. The inquiry appears to discuss calculations made before the second edition, so if you were using API Recommended Practice 530, it must have been the first edition.

Since the later calculations were made based on the second edition (1978), they appear to yield a different result. This could be due to three factors:

1) The differences in the Rupture Allowable Stress between the first and second editions; 2) the differences in the design equations between the first and second editions and; 3) the use of the design procedure.

The second part of the answer concerns the application of the API Recommended Practice 530 design procedure. The tubes covered in API Recommended Practice 530 are designed to operate at 540°C or higher which, for P 22, is the rupture range. When a tube is designed for operation in the rupture range, the design procedure adds a fraction of the corrosion allowance to the stress thickness when calculating the minimum thickness. This fraction is the corrosion fraction shown in Figure 1 of API Recommended Practice 530.

The calculation of the design metal temperature included with the inquiry shows that the full corrosion allowance was subtracted from the minimum thickness to calculate the stress thickness. If, however, only a fraction of the corrosion allowance were subtracted from the minimum thickness, then the calculated stress thickness would be larger. A larger stress thickness would yield a higher design metal temperature.

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API TITLE*1995 INT ** 0732290 0550512 390

One way to check the design using the current design procedure in API Recommended Practice 530 would be to calculate the minimum thickness for a range of design temperatures, using the appropriate corrosion fraction at each step, and find the design temperature that gives a minimum thickness equal to that of the original design.

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API TITLE*1995 INT ** 0732290 0550513 227

Issue Date: April 15, 1994

Publication: API Standard 598, *Valve Inspection and Testing*

Edition/Date: Sixth Edition, September 1990

Question: What valves are required to be tested in the vertical upright position during the low pressure closure test?

Reply: Valves required to be tested in the vertical upright position during the low pressure closure test are double block-and-bleed valves, valves other than double block-and-bleed which are having both seats tested simultaneously and gate valves 2 NPS and smaller which are having both seats tested simultaneously. For other than the above valves, the test position of the valve during the low pressure closure test is not defined by the standard.

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API TITLE*1995 INT ** ■ 0732290 0550514 163 ■

Issue Date: July 17, 1995

Publication: API Standard 600, *Steel Gate Valves—Flanged and Butt-Welding Ends*

Edition/Date: Ninth Edition, April 1991

Question: What is the definition of renewable and nonrenewable seats?

Reply: API Standard 600 does not address renewable or nonrenewable seats.

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API TITLE*1995 INT ** ■ 0732290 0550515 0TT ■

Issue Date: November 15, 1994

Publication: API Standard 600, *Steel Gate Valves—Flanged and Butt-Welding Ends*

Edition/Date: Ninth Edition, April 1991

Question 1: Is the flux core arc welding process permitted for weld overlay deposition on body and gate seat surfaces per Table 3?

Reply 1: API Standard 600 does not address the weld process that is to be used.

Question 2: Per Table 3, which weld overlay material is acceptable for trims No. 5 and No. 8 - R CoCr-A or E CoCR-A?

Reply 2: Either one may be used.

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API TITLE*1995 INT ** ■ 0732290 0550516 T36 ■

Issue Date: February 15, 1995

Publication: API Standard 607, *Fire Test for Soft-Seated Quarter-Turn Valves*

Edition/Date: Fourth Edition, May 1993

Question: Reference 4.4.8, does ASTM A 193, Grade B7 include ASTM A 193 Grade B7M?

Reply 1: No.

Question 2: Reference 4.4.8, does qualification of a valve with B7M fasteners also qualify valves with fasteners conforming to other ASTM specifications listed in Table 1 of ASME B16.34, except ASTM 307, Grade B?

Reply 2: No.

Question 3: Reference 4.5.2, may valves qualified by fire testing as specified in the third edition of API Standard 607 be supplied before May 1996 regardless of when the fire testing is done (e.g. 1992, 1994, 1996)?

Reply 3: Yes.

Question 4: Reference 4.5.2, are fire tests done in accordance with API Standard 607, fourth edition, only valid for three years?

Reply 4: No.

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API TITLE*1995 INT ** ■ 0732290 0550517 972 ■

Issue Date: April 15, 1994

Publication: API Standard 610, *Centrifugal Pumps for General Refinery Service*

Edition/Date: Seventh Edition, February 1989

Question: Do piping and appurtenances for auxiliary systems on pumps also include piping to lube oil console or seal flush run down reservoir?

Reply: Yes. The pipe sizes from ½-inch to 1½-inch schedule 80 pipe is required because of past problems relating to smashing and bending of schedule 40 pipe in these size ranges. Schedule 80 is required.

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API TITLE*1995 INT ** 0732290 0550518 809

Issue Date: November 15, 1994

Publication: API Standard 610, *Centrifugal Pumps for Petroleum, Heavy Duty Chemical, and Gas Industry Service*

Edition/Date: Seventh Edition, February 1989

Question: Does API Standard 610 increase a pumps allowable external forces and moments when subjected to occasional loads imposed by earthquake and wind reactions?

Reply: No. API Standard 610 does not specifically address either seismic loading or wind reaction loading. It does however address all normally occurring loads for prudent design. Prudent piping design should take into account any significant wind loads and seismic loads particularly for critical services.

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API TITLE*1995 INT ** ■ 0732290 0550519 745 ■

Issue Date: April 15, 1994

Publication: API Standard 613, *Special-Purpose Gear Units for Petroleum, Chemical, and Gas Industry Services*

Edition/Date: Third Edition, April 1988

Question 1: Reference 2.6.1.5, should gear tooth stiffness be considered in this calculation?

Reply 1: While it is understood that gear tooth stiffness may influence the lateral critical analysis, there currently is not an industry-wide accepted algorithm or mathematical model available that adequately deals with these influences. Thus, the gear tooth stiffness is not considered.

Question 2: Reference 2.6.2, does "static loading" include gear forces?

Reply 2: Only static loading is used. Gear forces cannot be duplicated on the balance machine and therefore, are not included.

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API TITLE*1995 INT ** 0732290 0550520 467

Issue Date: November 15, 1994

Publication: API Standard 617, *Centrifugal Compressors for Petroleum, Chemical, and Gas Service Industries*

Edition/Date: Fifth Edition, April 1988

Question: Refer to Section 2, 2.6.1 of Standard 617, Fifth Edition, which indicates compressor shafts shall be made of "one piece." Please define "one piece" and comment on whether a stacked, built up rotor construction with dowel pin torque transmission (sketch was furnished) can meet this API requirement?

Reply: The one piece requirement is defined as one integral complete shaft. The individual pieces as shown in the sketch do not constitute an integral "one piece" shaft, and therefore does not meet the intent of Standard 617, Paragraph 2.6.1.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1995 INT ** 0732290 0550521 3T3

Issue Date: November 15, 1994

Publication: API Standard 617, *Centrifugal Compressors for Petroleum, Chemical, and Gas Service Industries*

Edition/Date: Fifth Edition, April 1988

Question: In API Standard 617, Section 2.9.5.5, is the phrase "unfilter vibration" synonymous with "overall vibration?"

Reply: Yes, the phrase "overall vibration" can be considered as synonymous with "unfiltered vibration."

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1995 INT ** 0732290 0550522 23T

Issue Date: April 15, 1994

Publication: API Standard 620, *Design and Construction of Large, Welded, Low-Pressure Storage Tanks*

Edition/Date: Eighth Edition, June 1990

Question: Does API Standard 620 allow the use of material less than 3/16-inch thick for suspended decks in double wall tanks?

Reply: No. Refer to Tables R-2 and R-5.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1995 INT ** 0732290 0550523 176

Issue Date: March 17, 1995

Publication: API Standard 620, *Design and Construction of Large, Welded, Low-Pressure Storage Tanks*

Edition/Date: Eighth Edition, June 1990

Question: Is the note at the end of Example 3 in F.1.3 of API Standard 620 correct?

Reply: No. In using the alternative approach by the application of Figure F-1, a check shall be made that the compressive stress does not exceed the compressive stress value bounded by the line OABC in Figure 3-1. In your case, it is 12,230 pounds per square inch. This correction will be included in the next edition of the standard.

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API TITLE*1995 INT ** 0732290 0550524 002

Issue Date: November 8, 1994

Publication: API Standard 650, *Welded Steel Tanks for Oil Storage*

Edition/Date: Ninth Edition, May 1993

Question 1: Does the weld space requirements in 3.7.2.3 of API Standard 650 apply if the reinforcing pads are on the inside of the wall?

Reply 1: Yes.

Question 2: Does API Standard 650 specify a minimum space between annular ring butt joints and shell vertical butt joints?

Reply 2: No.

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Issue Date: November 9, 1994

Publication: API Standard 650, *Welded Steel Tanks for Oil Storage*

Edition/Date: Ninth Edition, May 1993

Question 1: Does the insert plate in Figure A (of attachment) meet the weld spacing requirements of Section 3.7.3 of API Standard 650?

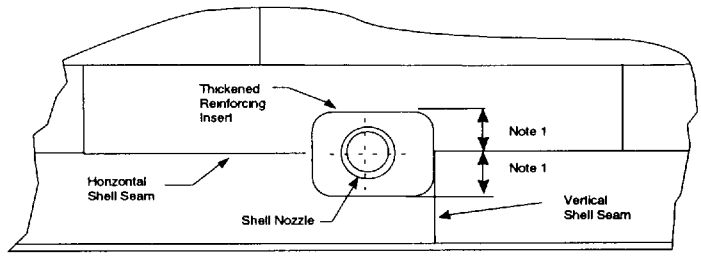
Reply 1: Yes.

Question 2: Does the insert plate in Figure B (of attachment) meet the weld spacing requirements of Section 3.7.3 of API Standard 650?

Reply 2: Yes.

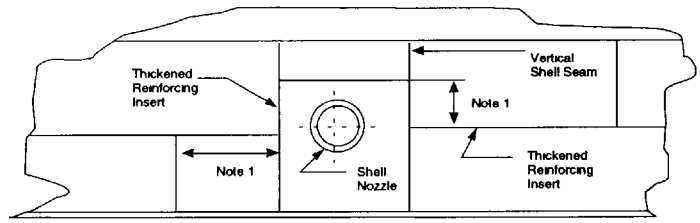
Question 3: Does the insert plate in Figure C (of attachment) meet the weld spacing requirements of Section 3.7.3 of API Standard 650?

Reply 3: Yes.



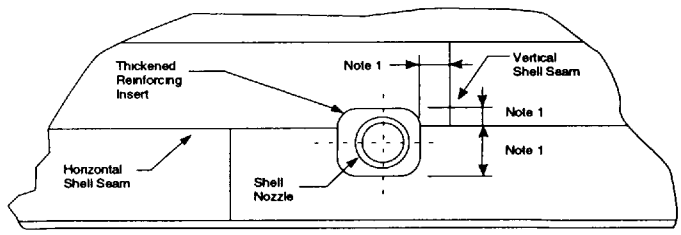
Note 1: Dimensions Comply with API 650. 3.7.3

Figure A



Note 1: Dimensions Comply with API 650. 3.7.3

Figure B



Note 1: Dimensions Comply with API 650. 3.7.3

Figure C

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API TITLE*1995 INT ** ■ 0732290 0550526 985 ■

Issue Date: November 9, 1994

Publication: API Standard 650, *Welded Steel Tanks for Oil Storage*

Edition/Date: Ninth Edition, May 1993

Question: Is it permissible to install stiffening rings on cone roof tanks to maintain the roundness, peaking and banding requirements set forth in Section 5?

Reply: No.

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API TITLE*1995 INT ** 0732290 0550527 811

Issue Date: June 23, 1995

Publication: API Standard 650, *Welded Steel Tanks for Oil Storage*

Edition/Date: Ninth Edition, May 1993

Question: Referring to API Standard 650, 5.2.3.6, is MT or PT required on any weld attaching: a) nozzles, manholes, and cleanout openings that are not in the stress relief assembly, b) attaching nozzles, manholes, and cleanout openings to a tank that stress relief is not a requirement, or c) attaching permanent and temporary attachments to a shell that are not classified in materials group IV, V, or VI?

Reply: a) Yes.

b) Yes.

c) No (see 5.2.3.5).

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API TITLE*1995 INT ** 0732290 0550528 758

Issue Date: September 15, 1995

Publication: API Standard 650, *Welded Steel Tanks for Oil Storage*

Edition/Date: Ninth Edition, May 1993

Question 1: Is the horizontal sweep board the same instrument as the vertical sweep board?

Reply 1: No.

Question 2: Is the horizontal sweep board a straight-edged instrument?

Reply 2: No.

Question 3: Is the vertical sweep board a straight-edged instrument?

Reply 3: Yes.

Question 4: Is it permissible to apply the horizontal (convex) or the vertical sweep boards to the inside diameter surface of the tank for dimensional evaluation of peaking or banding?

Reply 4: Yes.

Question 5: Is it permissible to apply the horizontal (concave) or the vertical sweep boards to the outside diameter surface of the tank for dimensional evaluation of peaking or banding?

Reply 5: Yes.

Question 6: Is the purpose of the sweep board to make contact with the tank in:

a) one place - (weld bead)?

b) two places - (weld bead and shell plate)?

c) two places - (shell plate on opposite sides of the weld jointly)?

Reply 6: a) and c) would satisfy the intent, but b) would not; c) is more accurate than a).

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API TITLE*1995 INT ** ■ 0732290 0550529 694 ■

Issue Date: November 20, 1995

Publication: API Standard 650, *Welded Steel Tanks for Oil Storage*

Edition/Date: Ninth Edition, May 1993

Question 1: Does API Standard 650 require a hydrostatic test before the tanks has been painted or primed (externally)?

Reply 1: No.

Question 2: If the tank is permitted to be primed externally before the hydrostatic test, is it mandatory to unprime the main welded joints on tank shells?

Reply 2: No.

Question 3: Is it mandatory to complete the entire tank erection, i.e., with floating roof, roof rolling ladder, roof seal, before the water test?

Reply 3: Yes. Refer to API Standard 650, 5.3.6.

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API TITLE*1995 INT ** ■ 0732290 0550530 306 ■

Issue Date: November 20, 1995

Publication: API Standard 650, *Welded Steel Tanks for Oil Storage*

Edition/Date: Ninth Edition, May 1993

Question 1: Does API Standard 650 require a hydrostatic test before the tanks has been painted or primed (externally)? If yes, is it mandatory to unprime or unpaint the welded seams of tank shell?

Reply 1: See API reply on previous page.

Question 2: What is the minimum distance between the fillet weld joining a stiffening ring to the tank shell and a main vertical seam?

Reply 2: This question will be referred to the appropriate subgroup for further action.

Question 3: Are there any tolerances applicable to the flatness of tank bottoms and roof decks?

Reply 3: No. Refer to API Standard 650, 5.2.2.1.

Question 4: For the nomenclature P.3 in Appendix P, what is the correct interpretation of "t", the corroded thickness of shell at the opening or the installation thickness? If circular reinforcing plates were used at the opening, what will be the correct value of "t"?

Reply 4: API Standard 650 does not specify which thickness should be used. For the question on reinforced openings, the shell thickness should be used.

Question 5: Since Appendix P is recommended for use with low-type nozzles, i.e., for L/2a - 1.0, why are Figures P-2D through P-2F and Figures P-2J and P-2L included in the appendix?

Reply 5: Refer to Paragraphs P.1 and 3.7.1.5, which do not preclude the application of Appendix P to nozzles located at L/2a only.

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API TITLE*1995 INT ** ■ 0732290 0550531 242 ■

Issue Date: November 20, 1995

Publication: API Standard 650, *Welded Steel Tanks for Oil Storage*

Edition/Date: Ninth Edition, May 1993

Question: Referring to 5.3.6 of API Standard 650, does the phrase "completion of the entire tank" allow the interpretation that the interior/exterior of the shell may be lined and/or coated prior to the water test?

Reply: The document does not specify when linings and/or coatings are to be applied in relation to testing.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1995 INT ** ■ 0732290 0550532 189 ■

Issue Date: March 17, 1995

Publication: API Standard 653, *Tank Inspection, Repair, Alteration, and Reconstruction*

Edition/Date: First Edition, January 1991

Question 1: Does API Standard 653 require that seismic loads be incorporated in tank shell evaluations?

Reply 1: Yes, see 2.3.3.4.

Question 2: For tanks constructed prior to the publication of API Standard 650, Appendix E, does API Standard 653 require that a seismic evaluation be conducted in accordance with API Standard 650, Appendix E?

Reply 2: No, the seismic evaluation referenced in API Standard 653, 2.3.3.4, does not have to be performed in accordance with API Standard 650, Appendix E, if the tank was constructed prior to the publication of Appendix E.

Question 3: What are the original construction standards for oil storage tanks constructed prior to 1965?

Reply 3: API Standard 653, Appendix A provided a list of documents that preceded the current edition of API Standard 650.

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API TITLE*1995 INT ** ■ 0732290 0550533 015 ■

Issue Date: November 20, 1995

Publication: API Standard 653, *Tank Inspection, Repair, Alteration, and Reconstruction*

Edition/Date: First Edition, January 1991

Question: Referring to API Standard 653, when shell penetrations are altered in accordance with 7.8.2.3, and all minimum weld spacing requirements are met, are the new welds re-joining the relocated plate (i.e. the plate removed with the penetration) to the shell plate required to be fully radiographed?

Reply: No. API Standard 653 does not currently address NDE on the welds covered in 7.8.2.3.

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API TITLE*1995 INT ** ■ 0732290 0550534 T51 ■

Issue Date: May 25, 1994

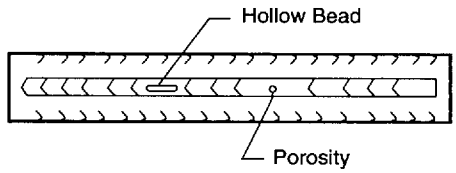
Publication: API Standard 1104, *Welding of Pipelines and Related Facilities*

Edition/Date: Seventeenth Edition, September 1988

Question 1: Please clarify the difference in rounded porosity and linear porosity as noted in API Standard 1104, 6.3.8.4. It is my interpretation that they are defined in 6.4.3.1, even though magnetic particle inspection is being referenced.

Reply 1: Rounded porosity is spherical in shape. Elongated porosity is round in cross section, but has length in the third dimension. Elongated linear porosity as used in 6.3.8.4 is elongated porosity whose third dimension is parallel to the weld. Hollow bead is elongated linear porosity that is in the stringer bead. Paragraph 6.4.1.3 is not the definition for linear porosity.

Question 2: Clarification is needed as to what hollow bead looks like. Does the diagram below correctly depict the appearance of hollow bead?



Reply 2: Yes, the diagram correctly depicts hollow bead.

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API TITLE*1995 INT ** 0732290 0550535 998

Issue Date: December 4, 1994

Publication: API Standard 1104, *Welding of Pipelines and Related Facilities*

Edition/Date: Seventeenth Edition, September 1988

Question 1: Is the weld procedure which uses a semi-automatic GMAW process for the root pass followed by a SAW process for the filler and cap passes, considered to be an automatic welding process as defined in Paragraph 1.2.2.11? The procedure utilizes a roll welding technique where the rotation of the weld is manually controlled. The rotating equipment is entirely independent of the welding equipment. The position of the sub-arc welding "gun" is manually adjusted as the multi-pass weld progresses.

Reply 1: No, the procedure you have described is a combination. The GMAW which you are using for the root pass is defined in 1.2.2.12, while the SAW process is defined in 1.2.2.11.

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API TITLE*1995 INT ** ■ 0732290 0550536 824 ■

Issue Date: December 8, 1994

Publication: API Standard 1104, *Welding of Pipelines and Related Facilities*

Edition/Date: Seventeenth Edition, September 1988

Question 1: The portion of 4.3, that is causing concern is the wording "When it is permissible to remove the lineup clamp before the root pass is completed..." Can you provide information that will assist in determining the conditions required for it to be permissible to remove the clamp?

Reply 1: The owner's requirements regarding the removal of line up clamps is discussed in 2.3.2.11 and it should be shown on the procedure specification form contained as Figure 1 on Page 4 of the standard. Here, under the heading "Type and Removal of Line Up Clamp" the owner indicates his requirement. This is what is referred to in the first sentence of 4.3. The 1104 Standard does not and can not establish when it is proper to remove a line up clamp. This must be done by the owner and recorded on the procedure specification form.

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API TITLE*1995 INT ** 0732290 0550537 760

Issue Date: April 4, 1995

Publication: API Standard 1104, *Welding of Pipelines and Related Facilities*

Edition/Date: Eighteenth Edition, May 1994

Question: Can a butt weld qualification test, as described in 2.6 of API Standard 1104, be used to support the weld specification for a fillet weld?

Reply: No. Paragraph 2.4.2.3 states that a major change in joint design is an essential variable. A change from a butt to fillet weld is a major change in joint design. Therefore, a butt weld qualification test does not qualify fillet welds.

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API TITLE*1995 INT ** 0732290 0550538 6T7

Issue Date: August 14, 1995

Publication: API Standard 1104, *Welding of Pipelines and Related Facilities*

Edition/Date: Eighteenth Edition, May 1994

Question 1: Why is it necessary to requalify a welding procedure when changing electrodes from Group 2 to Group 1, (specifically from E7010 to E8010) as required by 2.4.2.6a? Both electrodes are identical in every way except tensile strength; E8010 being l0ksi greater than E7010. Both electrodes have the same AWS specification (A5.5) and therefore have the same tolerance limits for chemical content. The last paragraph in 2.4.2.6 states that changes in filler metal may be made within the groups specified in 2.4.2.2, a and b. Considering that the mechanical properties of the weld would not be compromised by the new combination of filler metal to base metal, does this paragraph allow a change in electrodes without having to requalify the entire lprocedure, providing the base metal is the same as previously qualified?

Reply 1: This question has been discussed at the January 1995 meeting of the API-AGA Joint Committee on Oil and Gas Pipeline Field Welding Practices and it was agreed that there should be no change in the API Standard 1104. Items considered in this discussion included overmatching of weld to base metal strength, weld hardness, ductility and the many different alloys and combination of alloys that are permissible in an E8010 electrode.

Question 2: With respect to 2.3.2.3, Diameters and Wall Thicknesses, does the word "suggested" mean that one can expand or contract the range of a previously qualified procedure simply be revising the procedure to reflect the change? For example, if one wanted to weld 0.812 inch WT but had a procedure that only went up to 0.750 inch WT, could the WPS be revised without requalification?

Reply 2: It is left to the writer of the procedure specification to establish his own wall thickness and diameter ranges. Once the ranges have been established, the wall thickness cannot be changed to a wall thickness outside the range without requalifying the welding procedure because wall thickness is an essential variable as listed in 2.4.2.5.

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API TITLE*1995 INT ** 0732290 0550539 533

Issue Date: November 29, 1995

Publication: API Publication 1581, *Specifications and Qualification Procedures for Aviation Jet Filter/Separators*

Edition/Date: Third Edition, May 1989

Question 1: Question regarding building a filter separator that fulfills the requirements of Appendix A of API Publication 1581.

Reply 1: Appendix A of API Publication 1581 allows a manufacturer to certify a filter/separator vessel-element configuration as API qualified without having to physically test the configuration. Normally, element manufacturers test certain vessel-element configurations in accordance with Appendix A. You seem to be proposing a different approach, whereby you will certify qualification status by similarity to a configuration tested by another manufacturer. There is nothing in API Publication 1581 that prohibits this.

Also in your letter, you refer to constructing a vessel similar to a vessel that is part of an API qualified filter/separator vessel-element configuration. Your vessel would have to meet the criteria of Appendix A's Sections a and b in comparison with a vessel successfully qualified by another manufacturer's API Publication 1581 test. The elements you choose to use in your configuration would have to be the same model as the ones used by the other manufacturer in its test, and would also have to meet the requirements set out in Sections e, f and g of Appendix A. Finally, the internal layout of your proposed configuration would have to compare favorably with the selected test configuration in accordance with Sections h and i.

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API TITLE*1995 INT ** 0732290 0550540 255

Issue Date: April, 1994

Publication: API Standard 2000, *Venting Atmospheric and Low-Pressure Storage Tanks: Nonrefrigerated and Refrigerated*

Edition/Date: Fourth Edition, September 1992

Question 1: What formula should be used to calculate the theoretical flow for an untested manway cover which blows off during an overpressure?

Reply 1: The recommended method is described in 2.7.1.1.4 of API Standard 2000, fourth edition. This paragraph refers to the equation in 2.7.1.1.1.

Question 2: Does API Standard 650 contain a recommended practice for calculating vacuum ratings for tanks?

Reply 2: No. The scope of API Standard 650 includes tanks with internal pressures only. Although Standard 650 does not contain requirements for vacuum ratings, many companies size API 650 tank vents to prevent a vacuum inside the tank from exceeding -1 ounces per square inch. This is less than the minimum design roof live load of 25 pounds per square foot specified in section 3.10.2.1 of Standard 650. API Standard 620 does contain some information for vacuum design.

Question 3: How can vent size can be calculated after the vacuum rating is known?

Reply 3: Paragraph 2.4.2.1, "Inbreathing (Vacuum Relief)" and 2.6.1.2, "Vacuum" in API Standard 2000 present methods for determining the required vacuum flow for a tank. After the required flow is determined, capacity data from vent device manufacturers can be used to select the appropriate size. Section 2.7.1 of Standard 2000 gives several methods used by manufacturers to vent devices to determine the flow capacity.

Question 4: Will rules for vacuum relief allow us to use an untested vacuum relief device under the same 0.5 coefficient rule already described in Standard 2000?

Reply 4: No. Paragraph 2.7.1.1.4 allows the calculation method to be used for manhole covers only. Paragraph 2.7.2.2 requires vacuum relief devices to be stamped with a rated capacity. However, the capacity of an open vent can be determined using recognized flow equations, taking into account the configuration, presence of a screen, etc.

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API TITLE*1995 INT ** ■ 0732290 0550541 191 ■

Issue Date: March 22, 1995

Publication: API Standard 2000, *Venting Atmospheric and Low-Pressure Storage Tanks: Nonrefrigerated and Refrigerated*

Edition/Date: Fourth Edition, September 1992

Question 1: Does Paragraph 2.6.3.2 of API Standard 2000 require emergency vents from a tank located inside a building to discharge outside of the building?

Reply 1: Yes. Paragraph 2.6.3.2 of API Standard 2000 indicates that when a tank is located inside a building, the vents shall discharge to the outside of the building. This includes "normal vents" and "emergency vents."

Question 2: Is there a tank size below which emergency venting is no longer required?

Reply 2: No.

Question 3: Is there a tank size (rule of thumb) which a 2-inch vent provides adequate emergency venting capacity?

Reply 3: No. There are too many variables such as tank design pressure, latent heat of vaporization, etc., to provide a useable rule of thumb.

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API TITLE*1995 INT ** ■ 0732290 0550542 028 ■

Issue Date: April 12, 1995

Publication: API Standard 2000, *Venting Atmospheric and Low-Pressure Storage Tanks: Nonrefrigerated and Refrigerated*

Edition/Date: Fourth Edition, September 1992

Question: Clarification requested regarding API Standard 2000.

Reply: The task group reviewed your suggestion and decided not to change the current wording in API Standard 2000. The current wording generally discourages the use of F-factors that are less that one (rather than prohibiting their use). Please refer to Footnote "D" of Table 4 in API Standard 2000 for further explanation.

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API TITLE*1995 INT ** ■ 0732290 0550543 T64 ■

Issue Date: November 15, 1995

Publication: API Standard 2000, *Venting Atmospheric and Low-Pressure Storage Tanks: Nonrefrigerated and Refrigerated*

Edition/Date: Fourth Edition, September 1992

Question 1: How should thermal venting requirements for tanks larger than 180,000 barrels capacity be calculated?

Reply 1: Note 1 of Table 2 instructs the reader to calculate the venting rates for tanks greater than 180,000 barrels capacity since chart values for these size tanks are not included in Table 2. Notes A, B, and C present factors for calculating the required venting capacity.

Question 2: Can intermediate values of the environmental factors presented in Table 4 be obtained using interpolation?

Reply 2: Yes.

Question 3: Can the values in Table 2 be multiplied by 0.0283 to convert from SCFH (standard cubic feet per hour at 60°F) to SCMH (standard cubic meters per hour at 15.6°C)?

Reply 3: Yes.

Question 4: Can the values in Table 2 be used for tanks with an inert gas purge?

Reply 4: Yes. Also, regulator failures should be considered in vent sizing (see Paragraph 2.3.5.3).

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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